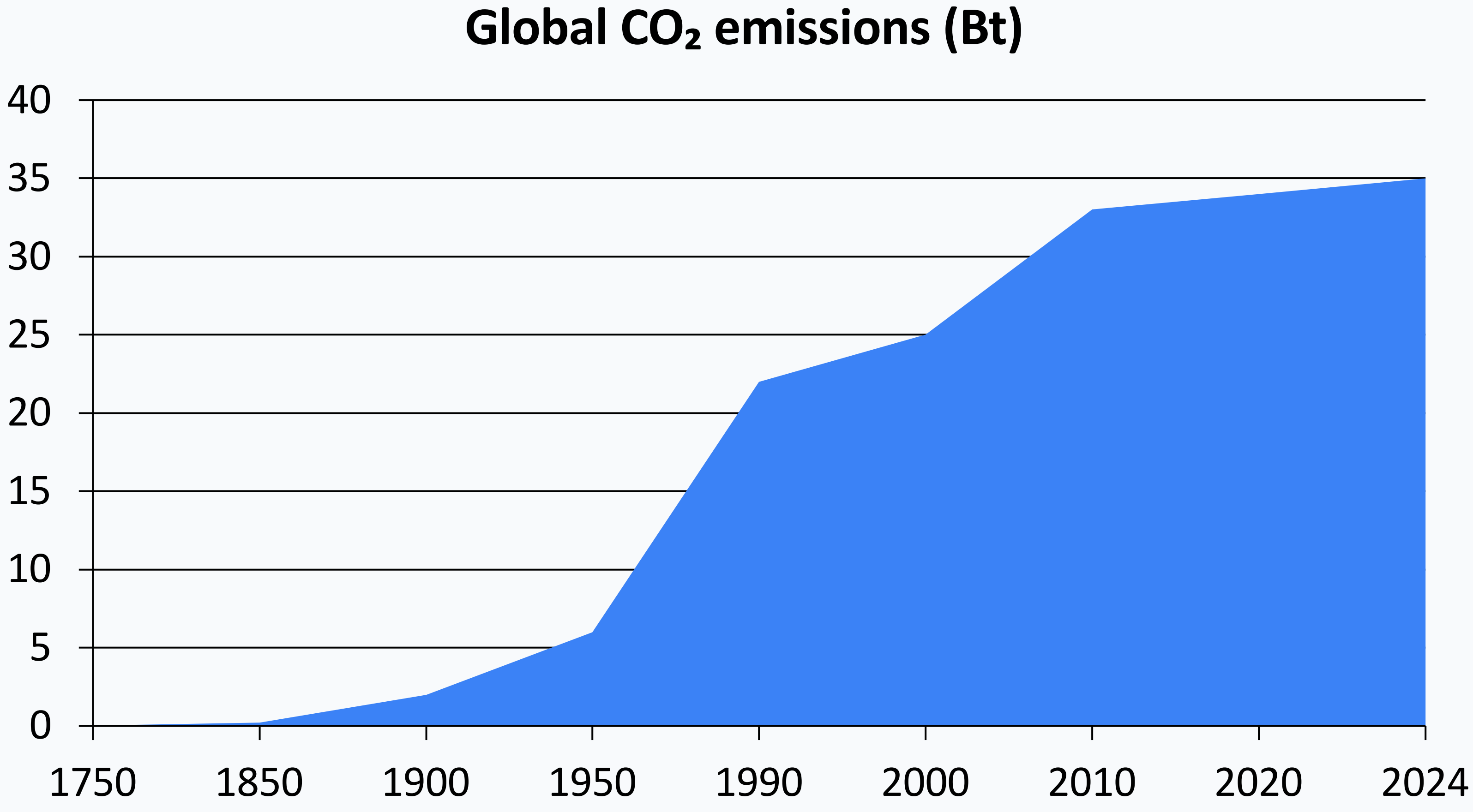
Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

For policy analysts, climate negotiators, and sustainability researchers

Data attribution: Global Carbon Budget 2025, as published by Our World in Data (Hannah Ritchie & Max Roser).

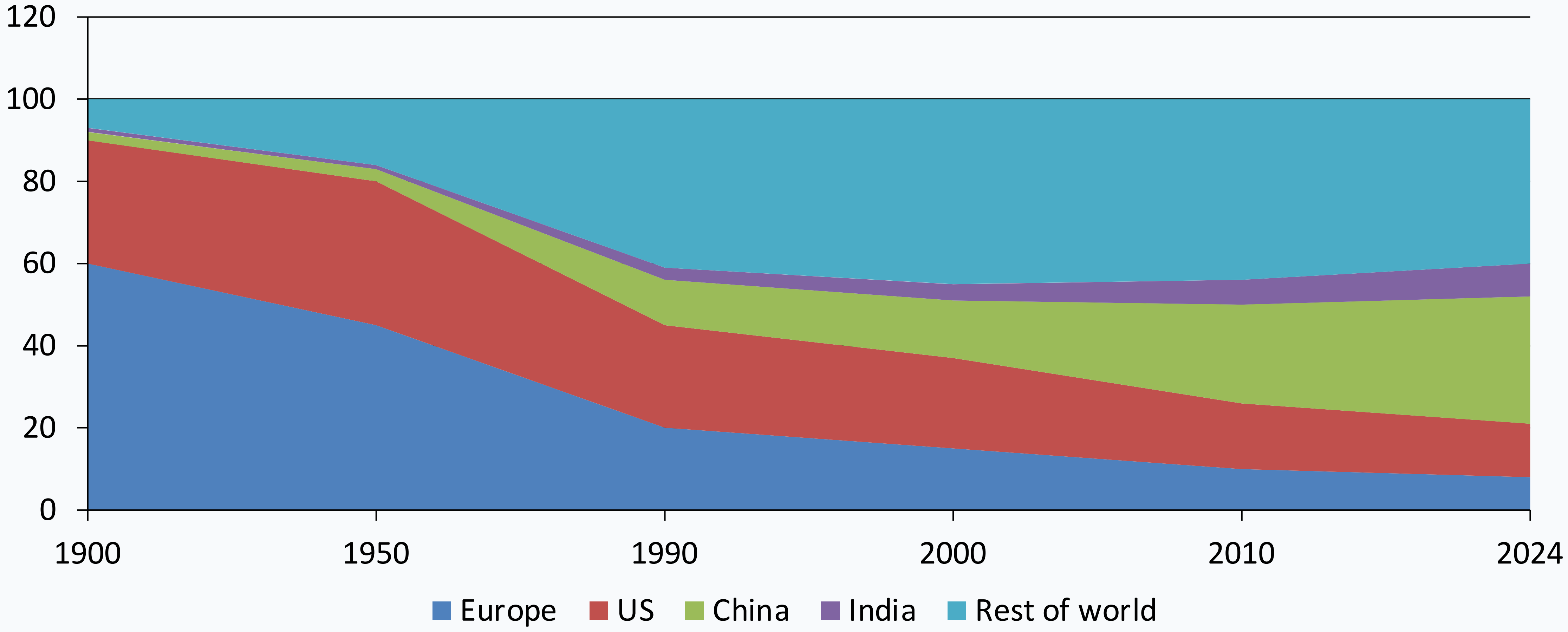
Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Inflection points: slow pre-industrial growth → post-1950 acceleration → slower growth, but no peak yet

Source: Global Carbon Budget (2025) via Our World in Data

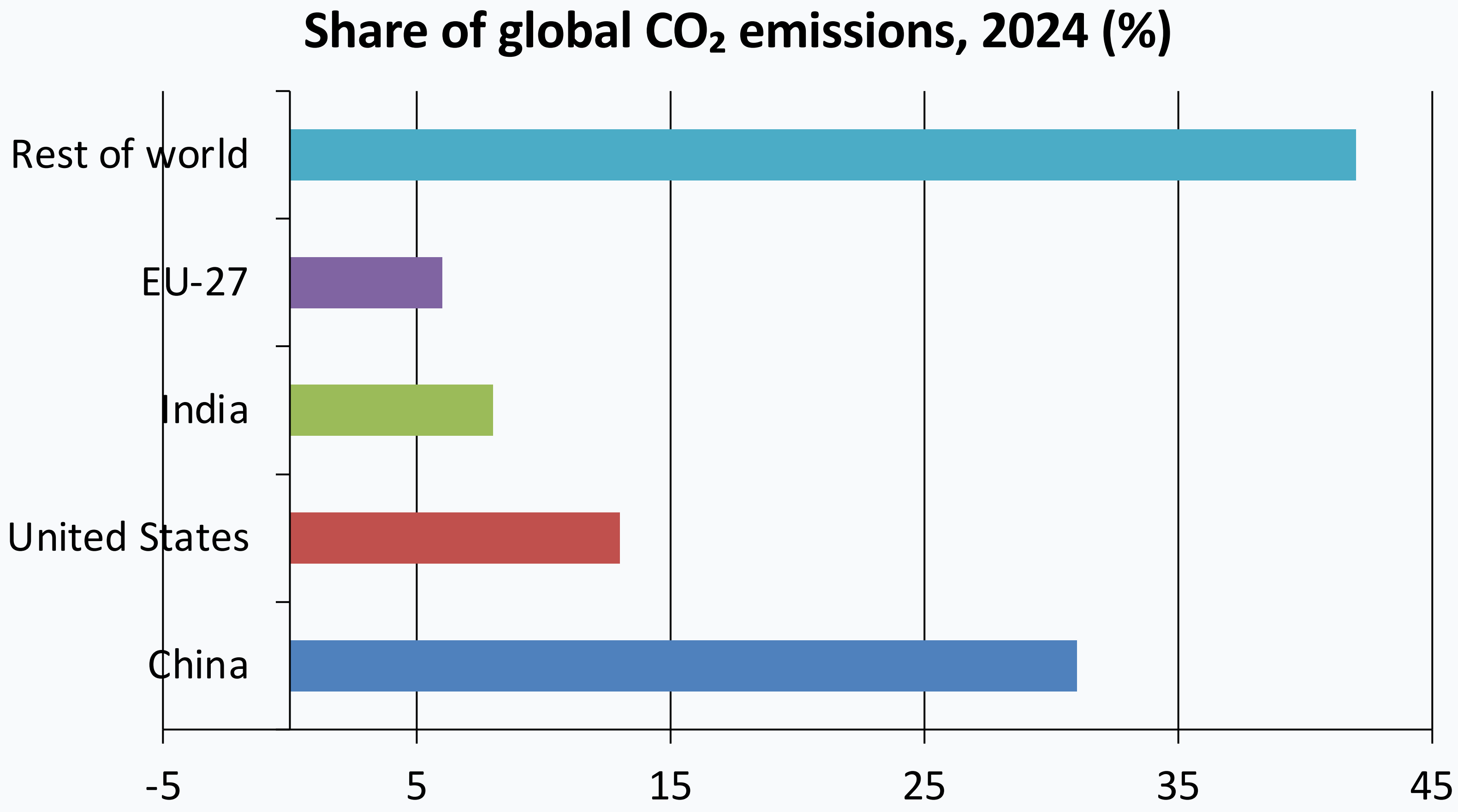
Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third



Regional share shift: historical dominance by Europe/US gave way to rapid growth in Asia, led by China

Source: Global Carbon Budget (2025) via Our World in Data

China Now Emits More Than One-Quarter of Global CO_2

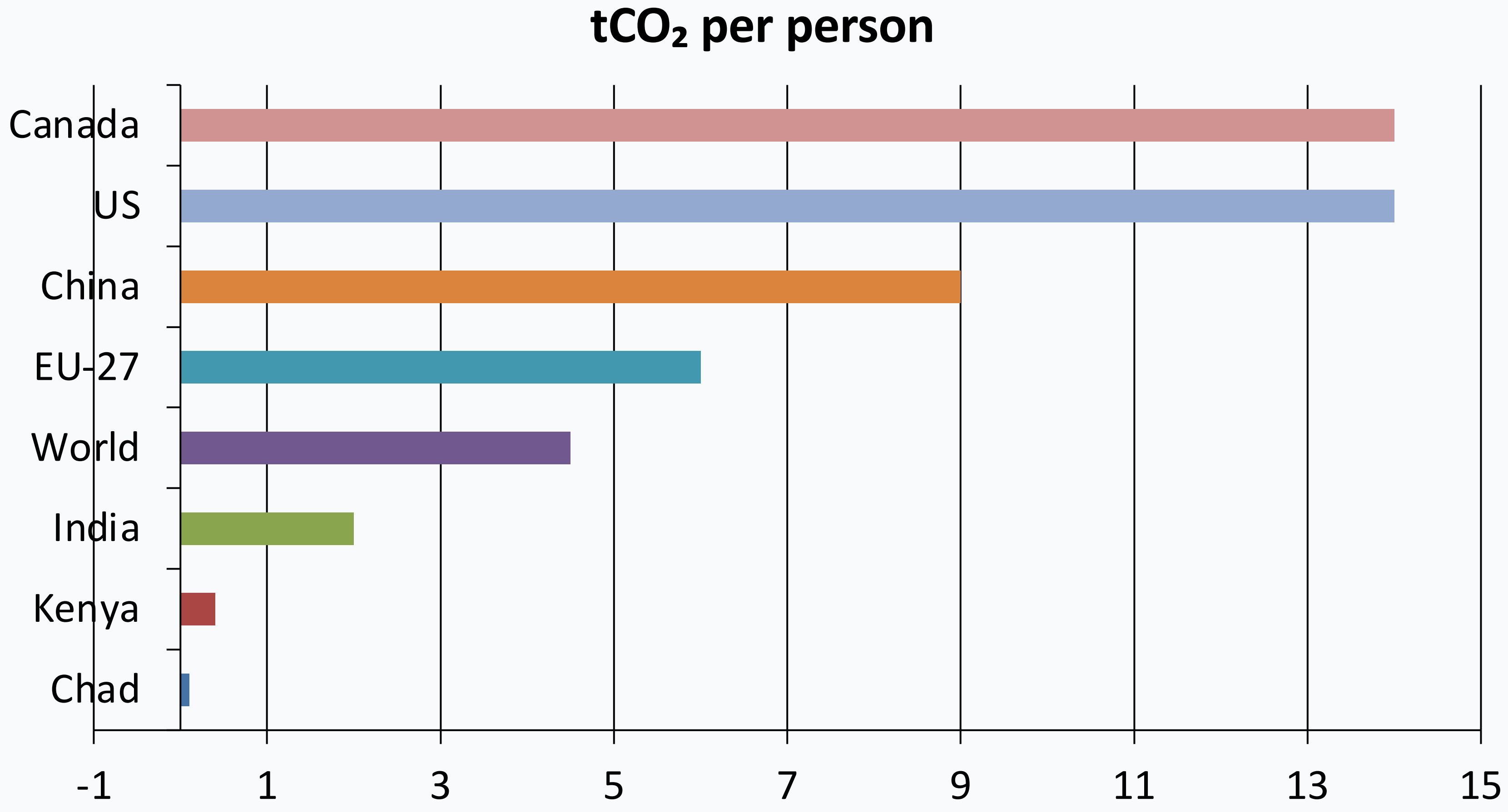


2024 shares

China: ~31%
US: ~13%
India: ~8%
EU-27: ~6%

Source: Global Carbon Budget (2025) via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Cana



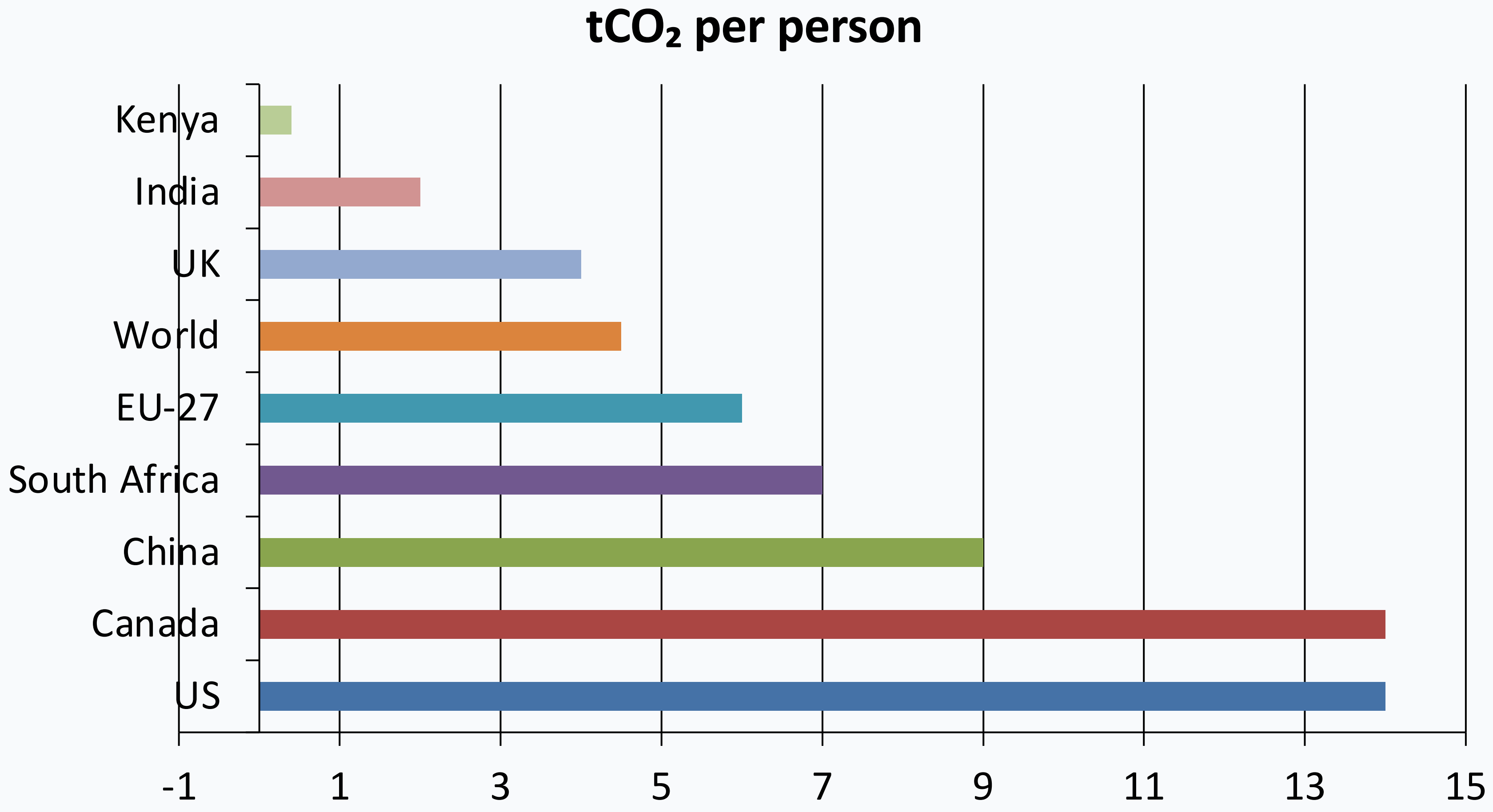
Inequality lens

~150x gap
0.1 t (Chad)
vs
~14 t (US/Canada)

France & UK are much lower than US despite similar living standards.

Source: Global Carbon Budget (2025) via Our World in Data

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2

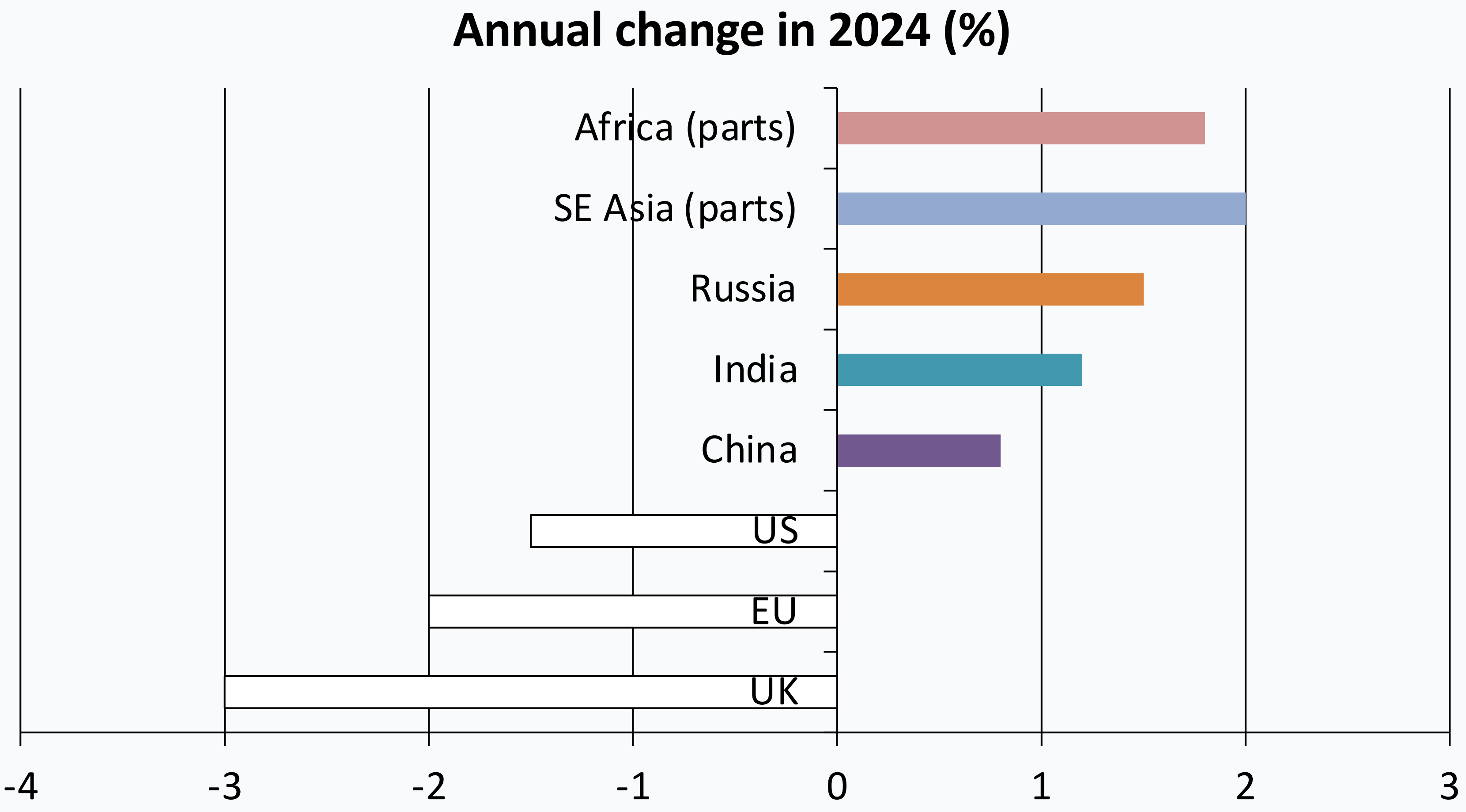


Context reference:
OWID figure_8.jpg

China is the largest total emitter, but still below US per-capita levels; India remains near ~2 t/person

Source: Global Carbon Budget (2025) via Our World in Data

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia

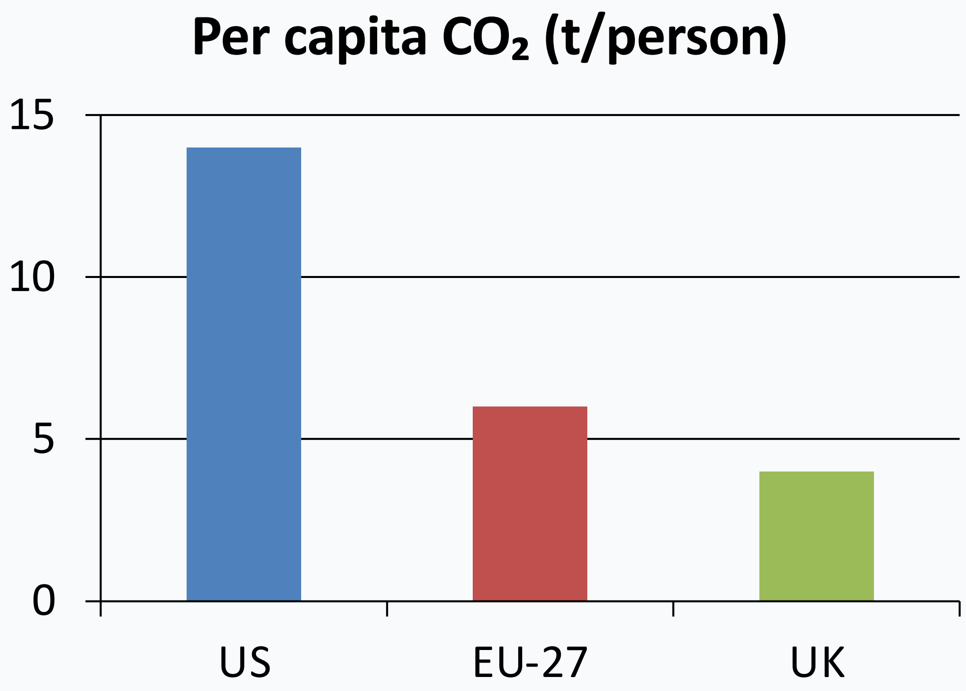


Diverging bar proxy
Blue: declines
Red: increases

Pattern: mixed global signal in 2024.

Source: Global Carbon Budget (2025) via Our World in Data

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than



Why the gap among wealthy countries?

• Similar prosperity does not imply similar emissions.
• France, UK, and Portugal have lower per-capita CO_2 than Germany, the Netherlands, or the US.
• A major driver is electricity mix: greater nuclear/renewables share tends to lower per-capita emissions relative to fossil-heavy systems.

Policy implication: energy system choices can decouple living standards from CO_2 intensity.

Source: Global Carbon Budget (2025) via Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

Global CO_2 emissions now exceed 35 Bt/year and have not yet peaked, despite slower growth.

China is the largest annual emitter, but its per-capita emissions remain below the US.

Historical responsibility was concentrated in Europe and the US, which dominated early emissions.

Per-capita gaps remain extreme: roughly 150x between lowest- and highest-emitting countries.

Energy policy and technology choices can materially reduce emissions without sacrificing prosperity.

Source: Global Carbon Budget (2025) via Our World in Data